Project Proposal: Text-Based Tariff Exposure & Event Stock Returns

Executive Summary and Target Audience

This project would give access to a predictive framework designed to quantify firm level tariff and supply chain vulnerability prior to trade policy shocks. By extracting natural language risk disclosures from the 10-k Fillings the project will generate exposures scores that will predict equity price falls after major trade policy announcements

Core Value Position

When trade policy shifts abruptly, it's hard for sectors to filter local producers from import dependent supply chains. By the end we will be able to utilize pre-event signals by processing disclosures before the market corrects the valuation.

Target Audience

1. Macro Strategist
2. Corporate Risk Officers
3. Capital Allocators

Observation Unit and Data Schema

- Unit of Observation: Firm_ticker

  The data set is structured from the observation of 1 stock, allowing a cross-sectional regression analysis and real time filtering to take place

Required Data Set Schema:

Column Name	Data Type	Description
ticker	String	Unique stock identifier (e.g., AAPL, WMT).
gics_sector	String	Global Industry Classification Standard sector assignment.
market_cap	Float	Firms Market Capitlization
tariff_exposure_score	Float	metric derived from 10-K filings that measures direct tariff exposure
import_dependency_score	Float	Quantitative metric capturing firm supply-chain reliance on overseas imports.
event_CAR_apr3_apr4	Float	Cumulative Abnormal Return measured across the April 3–April 4 event window.
pre_event_volatility	Float	Baseline daily return volatility calculated over the 60-day pre-event estimation window.

Sample Restrictions & Data Integrity

To preserve econometric validity and eliminate lookahead bias, strict sample selection rules are enforced:

a. All SEC edgar files will come out prior to april
b. Only using Equities from S&P500

Data Ingestion & Storage Architecture

- Project
  - Data
    - Raw
      - Edgar_Tarrifts
      - Markey_data

Storages Directory:

a. Edgar_Tariffs: Raw HTML+ Text files downloaded from the SEC EDGAR API before April 2025
b. Market_Data: Stores daily stock prices from Yahoo Finance alongside industry groups and standard financial benchmarks

Dashboard + Design:

X-axis: Tariff Exposure Score, Higher values indicate greater disclosed tariff risk
Y-axis: Cumulative Abnormal Return during the announcement window
Control Filters:

- GICS Sector Filter: Multi-select dropdown to isolate specific industry groups
- Market Capitalization Tiers

PORTFOLIO RISK DASHBOARD: TARIFF EXPOSURE vs. EVENT CAR (APRIL 3 - APRIL 4)

